

Simplex Problem-Solving Workshop

September 24, 2026

Name: _____

Ungraded practice

How to use this worksheet

Work through Problems 1–4 with the class. Problem 5 is optional practice. Write dictionaries as equations: solve for each basic variable, and substitute into every other row and the objective. For each ordinary pivot, record the entering variable, leaving variable, step length, and new objective value.

Conventions for every problem

All problems are **minimization** problems. All original, slack, surplus, and artificial variables are nonnegative.

- In a dictionary, a negative objective coefficient is an improving entering candidate. Only basic rows that decrease as that variable increases limit the step. If the row is $x_i = b_i + q_{ij}x_j + \cdots$ with $q_{ij} < 0$, its bound is $\theta \leq b_i/(-q_{ij})$.
- **Bland:** choose the smallest-index nonbasic variable with negative reduced cost. Among minimum-ratio ties, the smallest-index basic variable leaves. An artificial variable a comes after all numbered variables.
- **Most-negative reduced cost:** choose the most negative coefficient; break entering ties by smallest index and minimum-ratio ties by smallest basic index.
- **Lexicographic leaving rule (Problem 2):** write constraint rows with all variables on the left. Divide each eligible row by its positive entering-column coefficient and compare the vectors in the stated fixed order, starting with the RHS.
- **Phase I:** use Bland for ordinary pivots in these problems. Stop immediately when $w = 0$, since w is a sum of nonnegative artificial variables. If the optimum is positive, the original problem is infeasible.
- **After $w = 0$:** fix nonbasic artificials at zero. If an artificial remains basic at zero, solve its row for the smallest-index original nonbasic variable with a nonzero coefficient and substitute. Either coefficient sign is allowed in this zero-step exchange. Here “original” includes slack/surplus variables. If no such coefficient exists, the row reduces to $0 = 0$ and can be removed. Then remove artificial columns and restore the original objective.

Check each conclusion. Distinguish an optimal solution, an infeasible problem, and an objective unbounded below. A degenerate pivot changes the basis without changing the feasible point.

1 A bounded LP: most-negative reduced cost

Solve the following LP. Add slacks x_3, x_4, x_5 in the listed constraint order. Start with those three slacks basic and $x_1 = x_2 = 0$.

$$\begin{array}{ll}\min & f = -3x_1 - 5x_2 \\ \text{s.t.} & x_1 \leq 4, \quad x_2 \leq 6, \\ & 3x_1 + 2x_2 \leq 18, \quad x_1, x_2 \geq 0.\end{array}$$

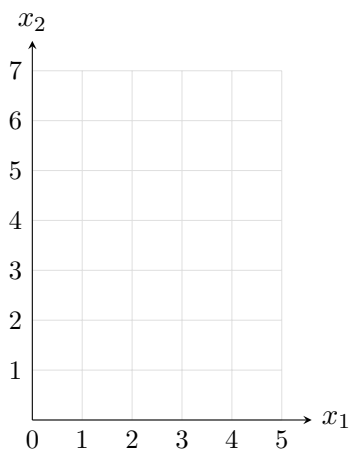
1. Write the initial dictionary, including the objective.
2. Use the **most-negative reduced-cost rule**. Write every dictionary until optimality, and give the final point and objective value.
3. Explain how the final objective row proves optimality. On the next page, plot the region and mark this path.

Work and explanation.

Problem 1 continued: compare Bland's rule

For the same LP, restart at the slack basis. Now use **Bland's rule**. Write the dictionaries, compare the visited points and number of pivots, and mark both paths on the graph. Does this example prove that one rule is always faster?

Work and explanation.



2 A zero step: Bland versus lexicographic leaving

Add slacks x_4, x_5, x_6 in constraint order and start at the slack basis.

$$\begin{array}{ll}\min & f = -x_1 \\ \text{s.t.} & x_1 - x_2 - x_3 \leq 0, \\ & x_1 - 2x_2 + x_3 \leq 0, \\ & x_2 + x_3 \leq 1, \quad x_1, x_2, x_3 \geq 0.\end{array}$$

1. Write the initial dictionary. Which variable enters, and which rows tie?
2. Compute **one pivot** using Bland's leaving rule.
3. Restart. Compute **one pivot** using the lexicographic leaving rule, with comparison order

$$(\text{RHS}, x_1, x_2, x_3, x_4, x_5, x_6).$$

4. Compare the bases, point, and objective. Do we need a positive step to change basis?

Work and explanation.

3 Two phases with one artificial variable

Add slack x_3 to the first constraint and subtract surplus x_4 from the second. Add an artificial variable a only to the second equality.

$$\begin{array}{ll}\min & f = -x_1 - 4x_2 \\ \text{s.t.} & x_1 + 2x_2 \leq 4, \\ & x_1 + 3x_2 \geq 4, \quad x_1, x_2 \geq 0.\end{array}$$

Phase I. Start with x_3, a basic, minimize $w = a$, and use Bland with order x_1, x_2, x_3, x_4, a .

1. Write the auxiliary equalities, starting dictionary, and objective after substitution.
2. Compute the first pivot. Show both ratios and resolve the tie explicitly.
3. Evaluate w . Has Phase I achieved its goal? Is a still basic?
4. Identify every original nonbasic variable that could replace a at this point. The next page completes the exchange and Phase II.

Work and explanation.

Problem 3 continued: remove the artificial and solve Phase II

At the end of Phase I, your dictionary should read

$$x_1 = 4 - 2x_2 - x_3, \quad a = -x_2 + x_3 + x_4.$$

1. Solve the artificial row for x_2 , substitute into the other row, and remove a . Show that the original point has not moved.
2. Restore $f = -x_1 - 4x_2$ by substitution. Use Bland for Phase II and write every dictionary.
3. Report the optimal point, value, and an objective-row certificate.

Work and explanation.

4 When Phase I proves infeasibility

Use slack x_3 , surplus x_4 , and one artificial variable a in the second constraint. Start with x_3, a basic and use Bland in Phase I.

$$\begin{array}{ll}\min & f = -x_1 - x_2 \\ \text{s.t.} & x_1 + x_2 \leq 1, \\ & x_1 + x_2 \geq 3, \quad x_1, x_2 \geq 0.\end{array}$$

1. Explain informally why the original constraints conflict.
2. Write the Phase I dictionary with $w = a$, and compute the first pivot.
3. Prove the resulting dictionary is optimal for Phase I. State w^* .
4. Should we restore the original objective and begin Phase II? Explain.

Work and explanation.

5 Optional: Phase II is unbounded

Subtract surplus x_3 and add one artificial variable a . Use Bland with order x_1, x_2, x_3, a .

$$\min \quad f = -x_1 - x_2 \quad \text{s.t.} \quad x_1 - x_2 \geq 1, \quad x_1, x_2 \geq 0.$$

1. Construct Phase I and find an original feasible basis.
2. Remove the artificial variable and restore the original objective.
3. Choose the next entering variable. Does any basic row limit its increase?
4. Write a feasible ray and calculate its objective as a function of the step. Compare this conclusion with Problem 4.

Work and explanation.